



Cardiac Arrest: Asystole & PEA

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA Tape.
- Focus on rapid and early BLS airway and ventilation tools. Intubation may not be the best option for these patients.
- Pediatric pads should be used in children < 10 Kg or PEDIA tape color purple.

Signs & Symptoms:

- Unresponsive
- Abnormal breathing (gasps)
- Pulseless
- Absent heart sounds
- Obvious death

Differential:

- Respiratory failure
- Foreign body airway obstruction
- Hyperkalemia
- Infection (Croup, epiglottitis)
- Hypovolemia
- Congenital heart disease
- Trauma
- Tension pneumothorax
- Hypothermia
- Toxins or Overdose
- Hypoglycemia
- Acidosis
- Acute MI or PE

Clinical Management Options

P	P	P	P	P	P	• Assess for unresponsiveness, absence of normal breathing, and pulselessness • Assess for obvious death criteria • Begin Pit Crew CPR procedure • BLS Airway Management and BVM with Oxygen as available • Passive oxygenation with nasal cannula at 25 LPM
L	L	L	L	L	L	• Airway management with iGel as needed • Monitor ETCO₂
1	2	3	4	5	6	• Vascular access IO can be placed first, but IV access is preferred when able to obtain. If patient ≤ 12 months, distal femur IO is preferred • Epinephrine • Fluid bolus with isotonic crystalloid as needed
						• Monitor and interpret ECG • Narrow PEA QRS ≤ 0.12 seconds: ◦ Consider mechanical causes - Cardiac tamponade, Tension pneumo, Mechanical hyperinflation, PE, Hypovolemia, Acute MI, Pump failure • Wide PEA QRS ≥ 0.12 seconds or Asystole: ◦ Consider metabolic causes - Tricyclic OD, Severe hyperkalemia, Acidosis, Calcium Channel Blocker OD, Acute MI, Pump failure. • Advanced Airway Management as needed. Intubation is not required if iGel is functioning appropriately with continuous waveform capnography. • Perform Needle Decompression for the asthmatic patient in arrest. • Perform Simple Thoracostomy for the asthmatic patient in arrest. • If ROSC then declare a resuscitation alert and use Post Resuscitation Checklist. • Targeted Temperature Management procedure if patient qualifies.

Consult Online Medical Control As Needed



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Pearls:

- Providers should not hasten departure after obtaining ROSC. Instead, focus on stabilizing the patient and ensuring that the airway is appropriately managed. If rearrest occurs, it will be in the first few minutes.
- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- In order to be successful in adult or pediatric arrests, a cause must be identified early and corrected. Resuscitation should include targeted therapies to address the underlying cause of the arrest.
- Respiratory arrest is a common cause of pediatric cardiac arrest. Unlike adults early oxygen and ventilation is critical.
- In most cases pediatric airways can be managed by basic interventions.
- Effective CPR is critical: 1) Push hard and fast at appropriate rate 2) Ensure full chest recoil 3) Minimize interruptions in CPR. Pause CPR < 10 seconds only.
- Effective CPR and treatment of underlying causes are the keys to successful resuscitation.
- Prolonged cardiac arrests may lead to tired providers and decreased compression quality. Ensure compressor rotation, summon additional resources as needed, and ensure provider rest and rehab during and post-event.
- For pediatrics use volume control device (IV Burette) for Dextrose and Fluid infusions
- Always quickly confirm asystole in more than one lead and, trouble shoot for Equipment settings/problems
- Reassess and document ETT/BIAD placement continuously after every move and at transfer of patient care.
- Continuous ETCO₂ should be initiated as soon as practicable.
- Calcium and sodium bicarbonate should be given early if hyperkalemia is suspected (renal failure, dialysis)
- Continue to use primary monitor for all event recording and data capture.
- All monitor event data and recordings are uploaded into e-PCR.
- Ultrasound to determine cardiac wall motion at pulse check; DO NOT interrupt compressions for ultrasound.